

PHASE 1: IDEATION PHASE

DEFINE PROBLEM STATEMENTS

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Team Size	4
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Project Name	HouseHunt: Finding Your Perfect Rental Home

1. PRIMARY PROBLEM STATEMENT

How might we create a secure, transparent, and efficient digital platform that connects property owners directly with potential renters while maintaining quality through administrative oversight, eliminating intermediaries and streamlining the entire property rental process?

2. STAKEHOLDER-SPECIFIC PROBLEMS

2.1 PROPERTY OWNERS

Problem Statement 1: Limited Market Reach Property owners struggle to reach their target audience effectively. Traditional methods like local advertisements, word-of-mouth, and broker networks have limited reach and high costs. There is no centralized platform where owners can list properties with complete information and reach a wide audience of potential renters.

Impact:

- Longer vacancy periods
- Lost rental income
- Dependence on expensive brokers
- Limited property visibility

Problem Statement 2: Verification & Trust Issues Property owners lack a reliable mechanism to verify renter credibility before renting out their properties. Concerns about property damage, rent defaults, and tenant behaviour create hesitation in the rental process.

Impact:

- Risk of property damage
- Potential rental payment defaults
- Time wasted on unreliable inquiries

- Legal complications

Problem Statement 3: Administrative Burden Managing property listings, responding to inquiries, scheduling viewings, handling paperwork, and collecting rent manually is time-consuming and inefficient. There is no integrated system to manage all these activities.

Impact:

- High time investment
- Missed booking opportunities
- Poor record keeping
- Difficult rent tracking

Problem Statement 4: Pricing Challenges Property owners lack access to market data and competitive pricing information, making it difficult to price their properties appropriately.

Impact:

- Overpricing leading to vacancy
- Under-pricing leading to revenue loss
- Competitive disadvantage
- Market misalignment

2.2 PROPERTY RENTERS

Problem Statement 1: Information Scarcity Property renters face significant challenges in obtaining complete, accurate, and transparent information about available properties. Current platforms either lack detailed information or provide misleading descriptions, leading to wasted time and frustration.

Impact:

- Multiple property visits required
- Wasted time and effort
- Poor rental decisions
- Unexpected costs

Problem Statement 2: Inefficient Search Process The property search process is fragmented across multiple platforms, classified ads, and personal networks. There is no single, comprehensive platform with advanced filtering capabilities to find suitable properties quickly.

Impact:

- Extended search duration
- Opportunity costs
- Decision fatigue
- Suboptimal choices

Problem Statement 3: Complex Booking Process The current booking process involves multiple intermediaries (brokers, agents), complicated paperwork, and lack of transparency. Direct communication with property owners is difficult, and the entire process is time-consuming.

Impact:

- High brokerage fees (10-15% of annual rent)
- Delayed bookings
- Lack of transparency
- Communication gaps

Problem Statement 4: Safety & Authenticity Concerns Renters face risks of fraudulent listings, fake properties, and unreliable landlords. There is no verification system to ensure property authenticity and owner credibility.

Impact:

- Monetary losses due to fraud
- Safety concerns
- Loss of trust in online platforms
- Stress and anxiety

2.3 PLATFORM ADMINISTRATORS

Problem Statement 1: Quality Control Maintaining platform quality and preventing spam, fake listings, and fraudulent activities is challenging without proper verification mechanisms. There is no systematic approach to vet property owners before they can list properties.

Impact:

- Platform reputation damage
- User trust erosion
- Legal liabilities
- Increased support requests

Problem Statement 2: User Management Complexity Managing multiple user types (Admin, Owner, Renter) with different access levels and requirements is complex. There is no robust role-based access control system.

Impact:

- Security vulnerabilities
- Unauthorized access
- Data breaches
- Administrative overhead

Problem Statement 3: Dispute Resolution Handling disputes between property owners and renters without a structured system leads to poor user experience and platform abandonment.

Impact:

- User dissatisfaction
 - Legal complications
 - Platform abandonment
 - Negative reviews
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3. TECHNICAL PROBLEMS

Problem 1: Authentication & Security

Statement: How to implement secure user authentication with role-based access control that prevents unauthorized access while maintaining user convenience?

Requirements:

- Secure password storage (encryption)
- JWT token-based authentication
- Role-based authorization
- Session management
- Protection against common attacks (XSS, CSRF, SQL injection)

Problem 2: Scalability

Statement: How to design a system architecture that can scale to handle thousands of users and properties without performance degradation?

Requirements:

- Efficient database queries
- Optimal API design
- Load handling capability
- Response time under 2 seconds
- Cloud-ready architecture

Problem 3: Data Management

Statement: How to design a database schema that efficiently handles relationships between users, properties, and bookings while maintaining data integrity?

Requirements:

- Normalized database design

- Efficient indexing
- Data validation
- CRUD operations
- Data backup and recovery

Problem 4: File Upload & Storage

Statement: How to handle multiple image uploads for properties with efficient storage and quick retrieval?

Requirements:

- Multiple image support
 - Image optimization
 - Secure file storage
 - Fast loading times
 - Storage management
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4. BUSINESS PROBLEMS

Problem 1: User Acquisition

Statement: How to attract both property owners and renters to the platform to achieve critical mass for network effects?

Challenge:

- Chicken-and-egg problem (need properties to attract renters, need renters to attract owners)
- Marketing budget constraints
- Competition from established platforms

Approach:

- Initial focus on quality over quantity
- Admin approval ensuring verified listings
- Word-of-mouth marketing
- Organic growth strategy

Problem 2: Trust Building

Statement: How to establish trust among users in a new platform where monetary transactions and property access are involved?

Challenge:

- New platform without established reputation

- User hesitation for online property transactions
- Security concerns

Approach:

- Admin verification system
- Transparent processes
- Clear privacy policies
- Responsive customer support

Problem 3: Monetization

Statement: How to generate revenue while keeping the platform affordable and competitive?

Challenge:

- Users resistant to additional fees
- Competition offering free listings
- Balancing revenue and user acquisition

Approach:

- Freemium model (MVP is free)
- Future premium features
- Value-added services
- Commission-based model for future

5. USER EXPERIENCE PROBLEMS

Problem 1: Complexity vs Simplicity

Statement: How to provide comprehensive features while maintaining interface simplicity for non-technical users?

Solution Approach:

- Clean, intuitive UI design
- Progressive disclosure of features
- Contextual help
- Minimal learning curve

Problem 2: Mobile Accessibility

Statement: How to ensure the platform is accessible and functional across devices (desktop, tablet, mobile)?

Solution Approach:

- Responsive design
- Mobile-first approach
- Touch-optimized interfaces
- Fast loading on mobile networks

Problem 3: Information Overload

Statement: How to present property information comprehensively without overwhelming users?

Solution Approach:

- Structured information display
- Visual hierarchy
- Progressive information reveal
- Smart defaults and filters

6. PROBLEM PRIORITIZATION MATRIX

Problem	Impact	Urgency	Feasibility	Priority
Authentication & Security	High	High	High	P0
Property CRUD Operations	High	High	High	P0
Admin Approval System	High	High	Medium	P0
Booking System	High	High	High	P0
Advanced Filtering	Medium	Medium	High	P1
Email Notifications	Medium	Low	Medium	P2
Payment Integration	High	Low	Medium	P3
Mobile App	Medium	Low	Low	P4

Priority Levels:

- **P0:** Must have (MVP)
- **P1:** Should have (Phase 2)
- **P2:** Could have (Phase 3)
- **P3:** Won't have initially (Future)
- **P4:** Long-term vision

7. SUCCESS CRITERIA

Measurable Outcomes

For Property Owners:

- Reduce property vacancy time by 50%
- Eliminate broker fees (10-15% savings)
- Reduce administrative time by 70%
- Increase property visibility by 300%

For Property Renters:

- Reduce property search time by 60%
- Access to 100% verified listings
- Save 10-15% on brokerage fees
- Complete transparency in process

For Platform:

- 99% uptime
 - <2s average response time
 - Zero security breaches
 - 90% user satisfaction rate
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8. CONSTRAINTS & LIMITATIONS

Technical Constraints

- Development timeline: 8 weeks
- Budget: Limited (student project)
- Team size: Individual developer
- Infrastructure: Free tier cloud services

Business Constraints

- No marketing budget
- No legal team
- Limited testing resources
- No customer support team

Scope Constraints

- MVP focus only
- No payment integration initially

- No mobile app in phase 1
 - Limited to rental properties only
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9. ASSUMPTIONS

1. Users have internet access and basic computer literacy
 2. Property owners are willing to provide accurate information
 3. Renters will use the platform ethically
 4. Admin can manually verify owners within 24 hours
 5. Cloud services remain available and affordable
 6. MongoDB Atlas free tier is sufficient for MVP
 7. Users accept terms and conditions
 8. Email is sufficient for initial communication
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10. RISKS & MITIGATION

Risk	Probability	Impact	Mitigation Strategy
Low user adoption	Medium	High	Quality over quantity, admin approval
Fraudulent listings	High	High	Admin verification, reporting system
Security breach	Low	Critical	Security best practices, regular audits
Performance issues	Medium	High	Optimized code, cloud scaling
Legal complications	Low	High	Clear T&C, disclaimer, consultation

11. CONCLUSION

The problem statements defined above provide a comprehensive understanding of:

1. **User Pain Points:** Clear identification of challenges faced by all stakeholders
2. **Technical Requirements:** Specific technical problems to solve
3. **Business Challenges:** Real-world business considerations
4. **Priority Framework:** Clear prioritization for development
5. **Success Metrics:** Measurable outcomes for evaluation
6. **Risk Awareness:** Identified risks with mitigation strategies

These problem statements will guide the solution design, development priorities, and feature implementation throughout the project lifecycle.

Next Steps:

- Create empathy maps for each user persona
- Define detailed requirements based on these problems
- Design solutions addressing each problem statement
- Develop implementation roadmap

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Part of House Hunt Documentation Package - Smart Bridge Virtual Internship